

UNIVERSITY OF MUMBAI



MASTER OF COMPUTER APPLICATION (MCA)

**In-Semester Capstone
Project Report
On
TIMSCDR Portal**

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Abstract

The rapid growth of digital technologies has significantly transformed the education sector, enabling institutions to manage academic activities efficiently through online platforms. Traditional methods of managing attendance, assignments, and student records are often manual, time-consuming, and prone to errors. To overcome these limitations, the **TIMSCDR Portal** has been developed as a web-based student management system that integrates multiple academic functionalities into a single platform.

The system provides role-based access for students, teachers, and administrators. Students can log in to view attendance records, access study materials, and mark attendance using QR-based technology. Teachers can manage academic resources and monitor student performance, while administrators have full control over managing classes, sessions, students, and teachers. The integration of QR-based attendance ensures accuracy, reduces proxy attendance, and improves efficiency.

By digitizing academic processes, the TIMSCDR Portal minimizes manual effort, enhances data accuracy, and improves communication among stakeholders. The system provides a user-friendly interface, secure authentication, and real-time access to academic data. It also lays the foundation for future enhancements such as mobile applications, cloud deployment, and AI-based analytics.

Acknowledgement

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We are also thankful to the **Head of the Department, Master of Computer Applications (MCA), University of Mumbai**, for providing a supportive academic environment and the necessary resources required to carry out this project effectively.

We extend our heartfelt thanks to all the faculty members of the MCA department for their guidance, knowledge, and support during the course of this project. Their valuable teachings and suggestions greatly contributed to the successful completion of this work.

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Chapter 1

1.1 Introduction

In modern educational institutions, managing student-related activities such as attendance, academic records, and learning materials is a complex and continuous process. Traditional systems rely heavily on manual record-keeping, which can lead to data inconsistency, inefficiency, and increased administrative workload.

The TIMSCDR Portal is designed to address these challenges by providing a centralized and automated system for managing academic operations. It integrates various modules such as student login, teacher login, admin dashboard, attendance tracking, and study material management into a single web-based platform.

The system ensures that all users can access relevant information based on their roles. Students can easily track their academic progress, teachers can efficiently manage classroom activities, and administrators can oversee the entire system. The portal enhances productivity, reduces paperwork, and provides a seamless digital experience.

1.2 Problem Statement

Educational institutions often face several issues while managing academic data manually. Attendance tracking is time-consuming and susceptible to manipulation. There is no centralized platform for storing and accessing study materials. Communication between students, teachers, and administrators is often inefficient. These challenges lead to poor data management, lack of transparency, and reduced productivity.

The TIMSCDR Portal aims to solve these problems by providing an automated, secure, and user-friendly system that streamlines academic processes.

1.3 Objectives

The main objective of the TIMSCDR Portal is to develop a comprehensive system that simplifies academic management. The system aims to automate attendance using QR codes, provide secure login for different users, manage academic records efficiently, and enable easy access to study materials.

Another important objective is to reduce manual work, improve accuracy, and enhance the overall user experience through a modern and responsive interface.

1.4 Scope of the Project

The TIMSCDR Portal is designed for use in schools and colleges to manage academic activities efficiently. It is a web-based application that can be accessed from any device with internet connectivity. The system can be further extended to include mobile applications, cloud-based deployment, and advanced analytics features.

Chapter 2

2. SYSTEM STUDY

System study is an important phase of software development in which the existing system is analyzed and the requirements of the proposed system are clearly understood. This phase helps in identifying system needs, planning development activities, and selecting the appropriate technologies for implementation. The system study for the **TIMSCDR Portal (Student Management System with QR Attendance)** is discussed below.

2.1 Requirement Analysis

Requirement analysis is the process of identifying what the system should do and how it should perform. In this project, requirements are gathered by analyzing the needs of students, teachers, and administrators.

From the student's point of view, the system should provide a simple and secure login facility. After logging in, students should be able to mark attendance using QR code scanning, view their attendance records, access study materials, and manage their profile information. The system should provide real-time updates and an easy-to-use interface for better user experience.

From the teacher's perspective, the system should allow secure login and provide features to manage academic activities. Teachers should be able to upload study materials, monitor student attendance, and maintain class-related data efficiently.

From the administrator's point of view, the system should provide a powerful dashboard with full control over system operations. The admin should be able to manage students, teachers, classes, class arms, sessions, and terms. The administrator should also be able to monitor attendance reports and ensure smooth functioning of the system.

Apart from functional requirements, the system must also satisfy non-functional requirements such as security, reliability, performance, and usability. The system should ensure secure authentication, fast response time, data accuracy, and a user-friendly interface. It should also be scalable for future enhancements.

2.2 Planning and Scheduling

Planning and scheduling play an important role in completing the project within the given time frame in a systematic manner. The development of the TIMSCDR Portal was divided into different phases to ensure smooth execution.

Initially, requirement gathering and analysis were carried out to understand system needs. After that, system design was created using diagrams such as DFD, ER diagram, and use case diagram. In the next phase, coding was performed using HTML, CSS, JavaScript, and PHP for different modules such as login, admin dashboard, student dashboard, and attendance system.

Once development was completed, testing was conducted to identify errors and ensure proper functioning of all modules. Finally, documentation and project presentation were prepared. This structured planning helped in completing the project efficiently and within the deadline.

2.3 Preliminary Product Description

The TIMSCDR Portal is a web-based student management system designed to simplify academic operations in educational institutions. The system provides a centralized platform where students, teachers, and administrators can perform their respective tasks efficiently.

The system includes features such as role-based login, QR-based attendance marking, study material management, student and teacher management, and attendance reporting. Students can mark attendance using QR codes and view their attendance statistics. Teachers can upload learning materials and monitor student activities. Administrators can manage all system components including classes, sessions, and users.

The system reduces manual work, improves accuracy, and enhances communication between users. It provides a modern and responsive interface that makes academic management more efficient and transparent.

2.4 Justification of Platform

The web platform was selected for developing the TIMSCDR Portal because it provides high accessibility and flexibility. A web-based system allows users to access the application from any device with an internet connection without installing additional software.

Web technologies such as HTML, CSS, JavaScript, PHP, and MySQL enable faster development, easy maintenance, and scalability. Since educational systems require real-time access and centralized data management, a web-based solution is the most suitable platform.

Additionally, the web platform reduces operational costs, supports multi-user access, and allows easy future enhancements such as cloud deployment and mobile integration.

2.5 Conceptual Model

The conceptual model of the TIMSCDR Portal explains how different components of the system interact with each other. In this system, users (students, teachers, and administrators) interact with the web interface to perform various operations.

Students log into the system and scan QR codes to mark attendance, view records, and access study materials. Teachers upload materials and monitor student activities. Administrators manage users, classes, sessions, and system data.

All user requests are processed by the backend system and stored in the database. The database maintains records of students, teachers, attendance, classes, and study materials. The system ensures smooth data flow between users and the database, providing accurate and real-time information.

Chapter 3

3. ANALYSIS AND DESIGN

The analysis and design phase is a crucial part of the software development life cycle. In this phase, the requirements identified during the system study are transformed into a structured system design. This helps in defining the system architecture, data flow, database structure, and overall functionality of the application before actual implementation.

For the **TIMSCDR Portal (Student Management System with QR Attendance)**, this phase ensures that all modules such as student management, teacher management, attendance system, and study material handling are properly designed and integrated. A well-defined design improves system efficiency, scalability, and maintainability.

3.1 Hardware Requirements

The following hardware components are required for the development and execution of the TIMSCDR Portal:

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB (8 GB recommended for better performance)
- **Hard Disk:** Minimum 100 GB free space
- **Internet Connection:** Required for accessing the web application and future deployment
- **Input Devices:** Keyboard and Mouse
- **Output Devices:** Monitor

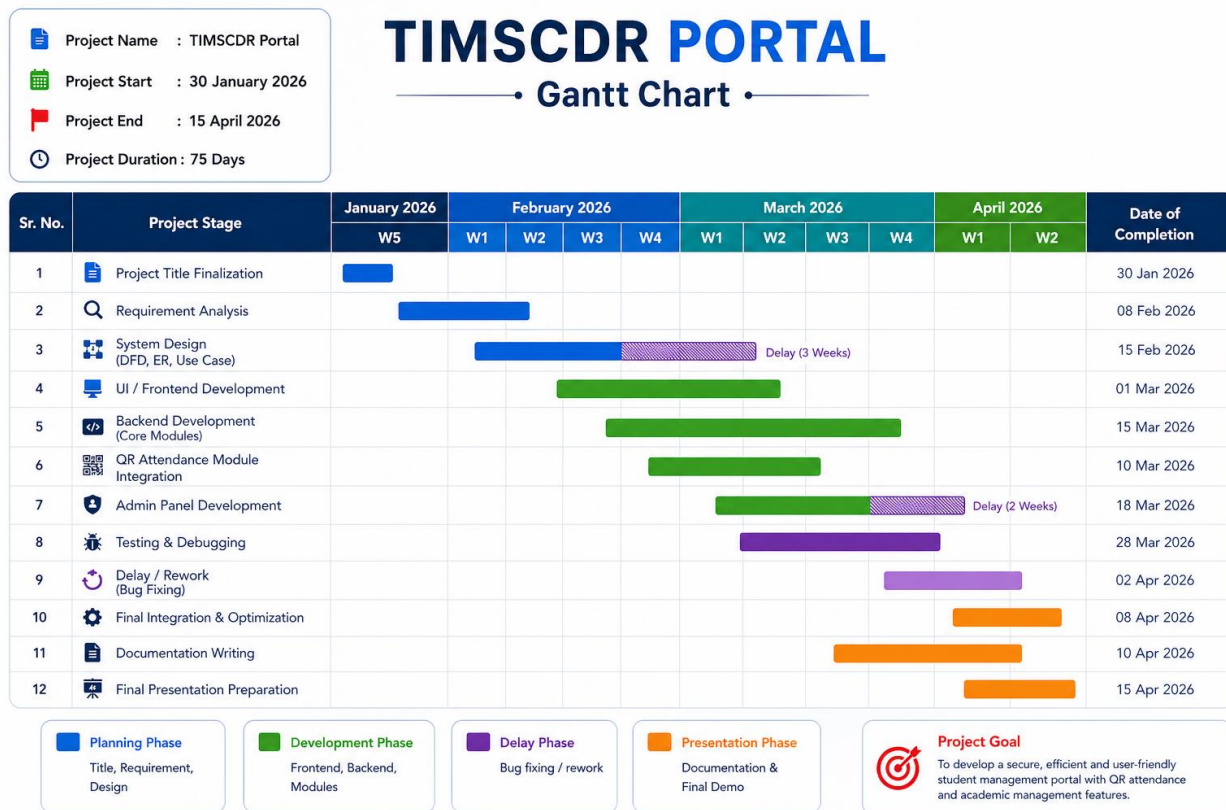
3.2 Software Requirements

The software requirements for developing and running the TIMSCDR Portal are as follows:

- **Operating System:** Windows 10 / Linux / macOS
- **Frontend Technologies:** HTML, CSS, JavaScript
- **Backend Technology:** PHP
- **Database:** MySQL
- **Web Server:** Apache (XAMPP/WAMP)
- **IDE/Editor:** Visual Studio Code
- **Browser:** Google Chrome / Mozilla Firefox

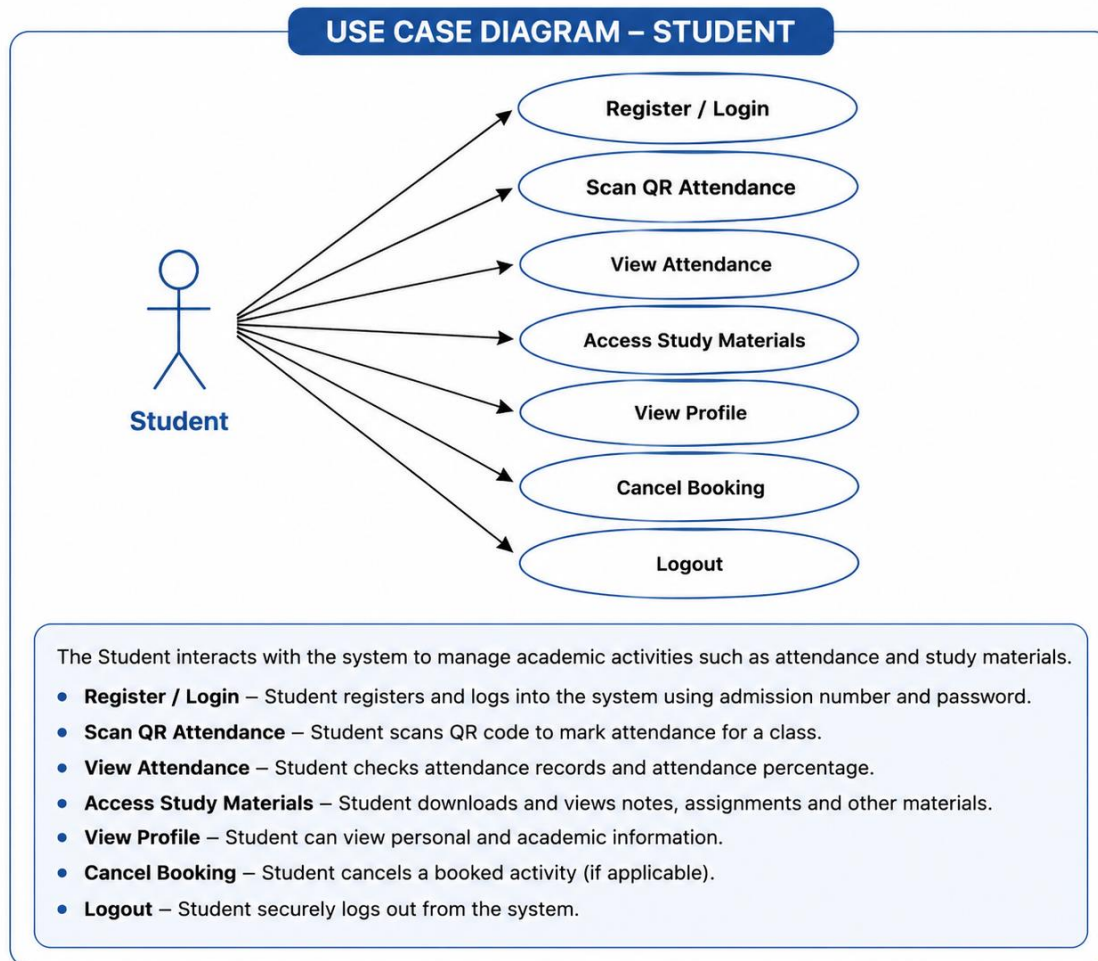
3.3 Gantt chart

The Gantt chart represents the timeline followed for the development of the TIMSCDR Portal. The project started with requirement analysis and system design, followed by frontend and backend development. The QR attendance module and admin panel were developed alongside core functionalities. Testing and debugging were conducted to ensure system reliability. A short delay period was included for bug fixing and improvements. Finally, documentation and presentation were prepared, leading to successful project completion on 15 April 2026.



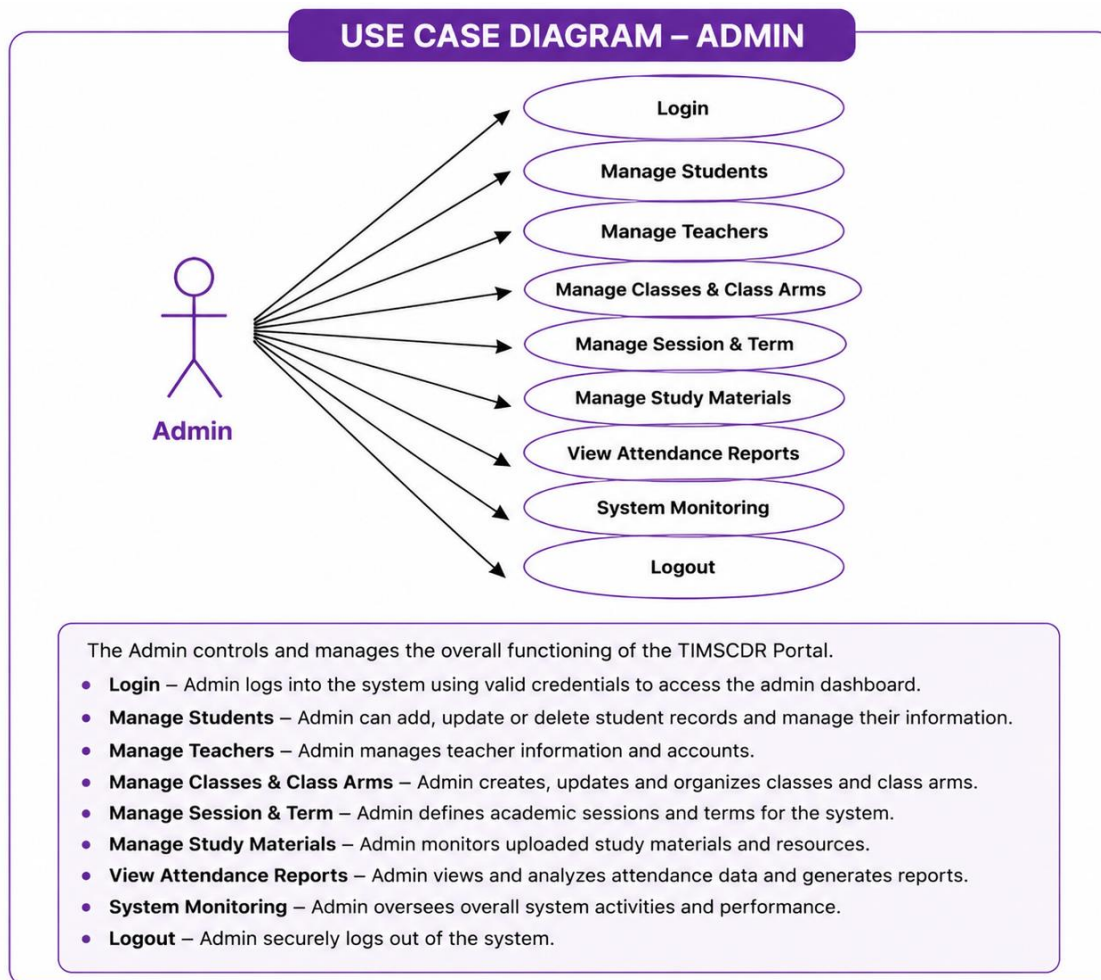
3.4 Use Case Diagram

The student interacts with the system to manage academic activities such as marking attendance using QR codes, viewing attendance records, and accessing study materials. The system provides a simple and secure interface for students to track their academic progress.



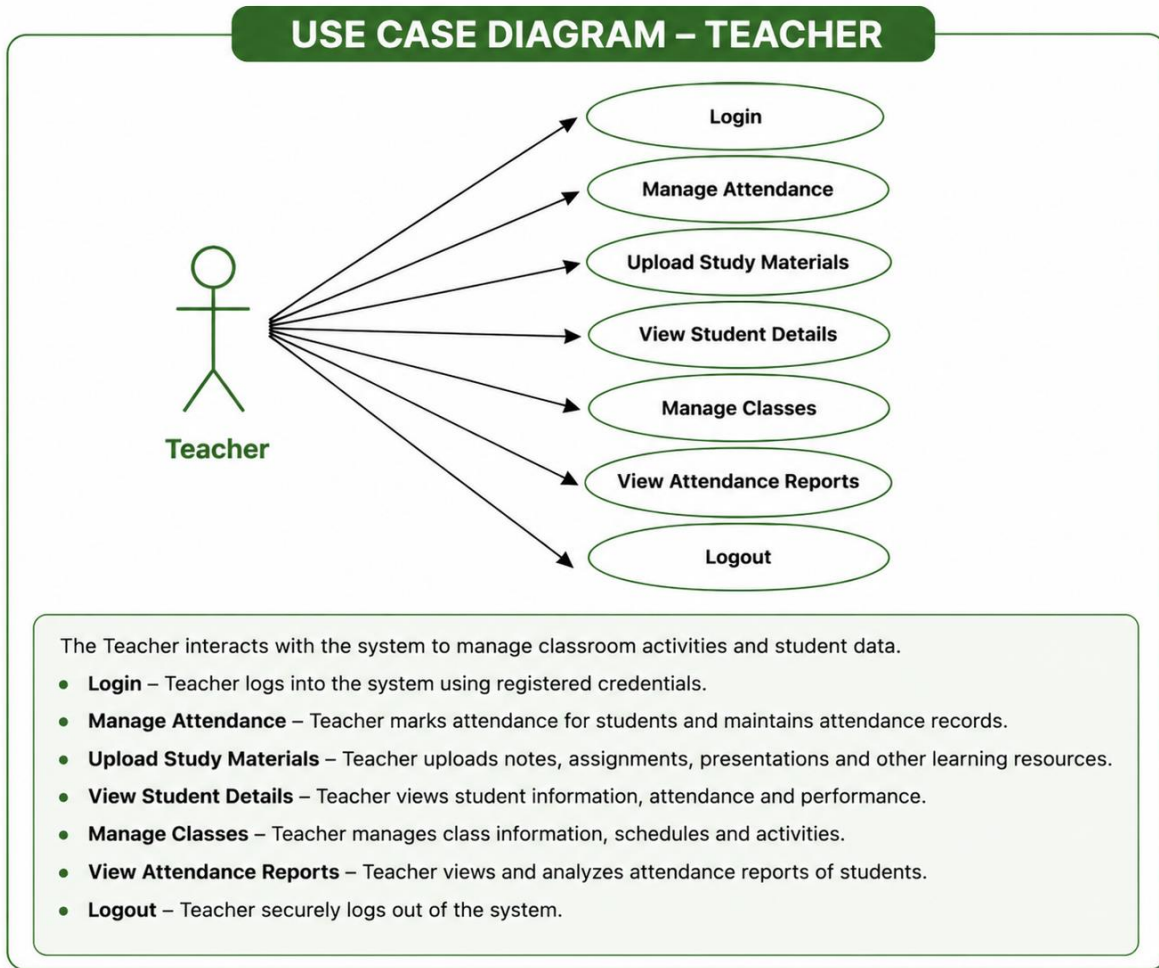
Use Case Diagram – Student

The student interacts with the system to manage academic activities such as marking attendance using QR codes, viewing attendance records, and accessing study materials. The system provides a simple and secure interface for students to track their academic progress.



Use Case Diagram – Admin

The admin controls and manages the overall functioning of the TIMSCDR Portal by handling students, teachers, classes, and academic sessions. The system provides a centralized dashboard for monitoring data and ensuring smooth operation.



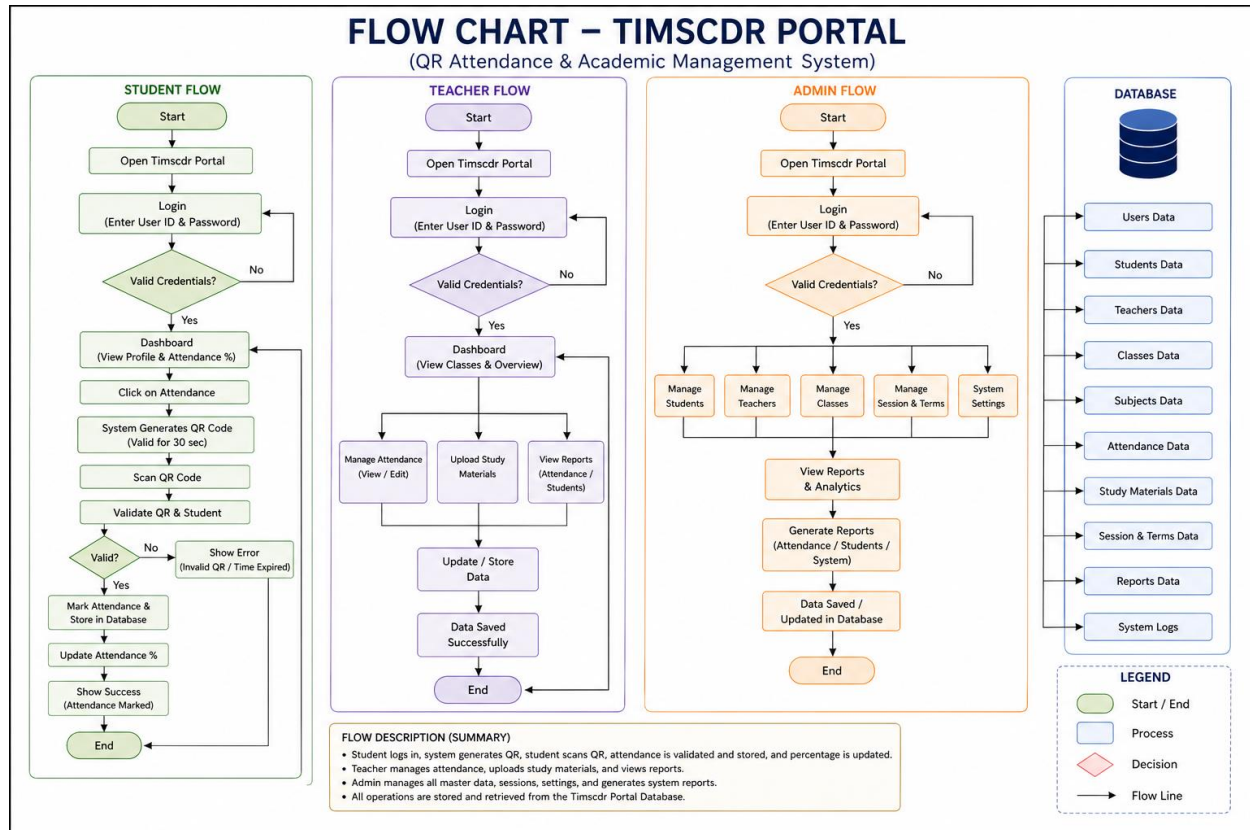
Use Case Diagram – Teacher

The teacher interacts with the system to manage classroom activities, including maintaining attendance records and uploading study materials. The system helps teachers efficiently monitor student performance and manage academic resources.

Flow Chart

The flowchart represents the step-by-step working of the TIMSCDR Portal system, showing how students, teachers and admins interact with the system to perform various operations.

It illustrates the complete process from login to data processing, storage, and output generation, ensuring smooth and efficient system functionality.



3.5 Context Diagram

The Context Diagram of the **TIMSCDR Portal** represents the system as a single process that interacts with three main external entities: **Student, Teacher, and Admin**.

The **Student** provides inputs such as login credentials and QR scan data, and receives outputs like attendance records, study materials, and profile details.

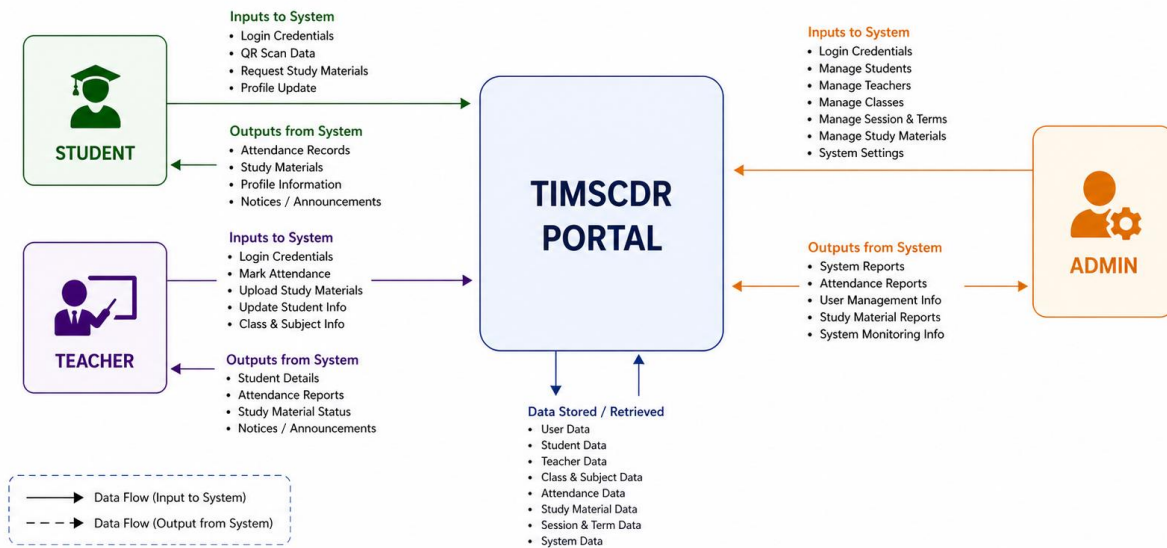
The **Teacher** interacts with the system by providing inputs such as attendance data and study materials, and receives outputs such as student information and attendance reports.

The **Admin** controls the system by managing students, teachers, classes, sessions, and study materials. The admin receives system reports and monitoring information as output.

All these interactions are processed by the central system (**TIMSCDR Portal**) and stored in the database, ensuring smooth data flow and system functionality.

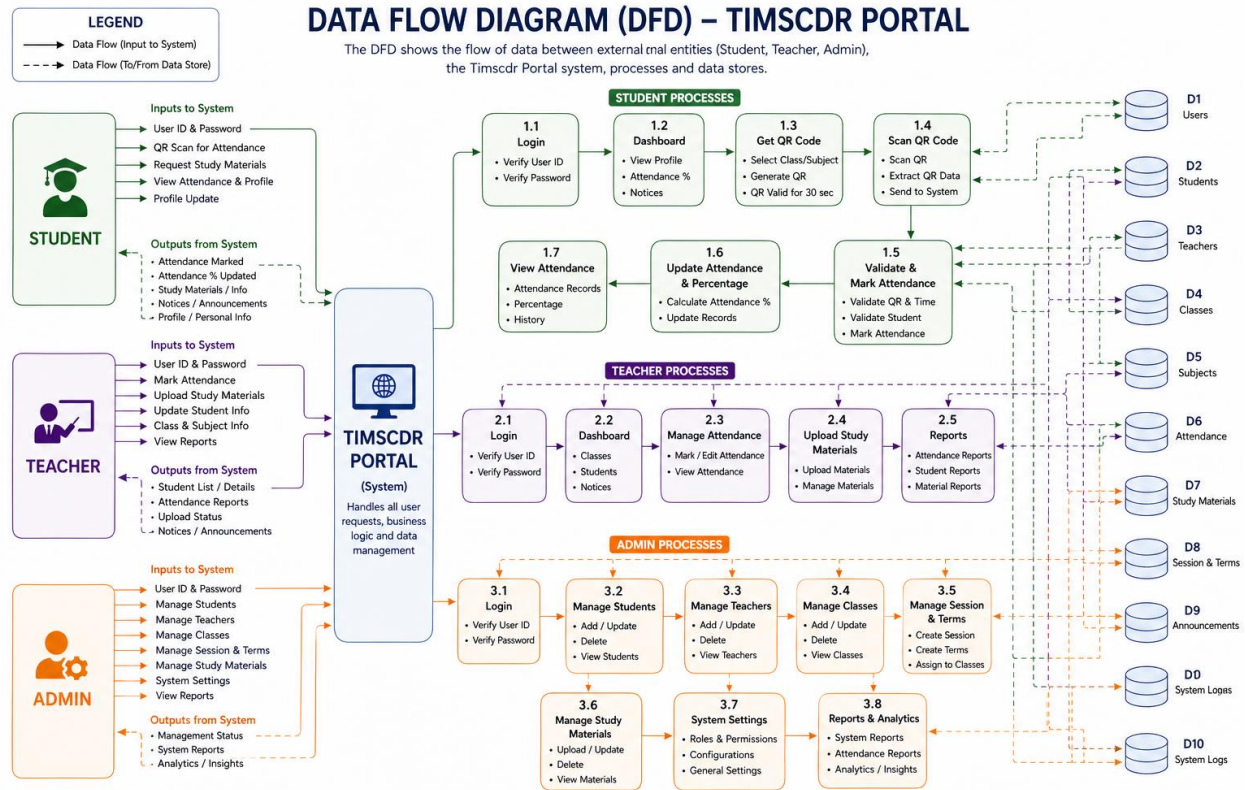
DATA FLOW DIAGRAM (DFD) – TIMSCDR PORTAL

The DFD represents the flow of data between external entities (Student, Teacher, Admin) and the Timscdr Portal system.



3.6 DFD level 1

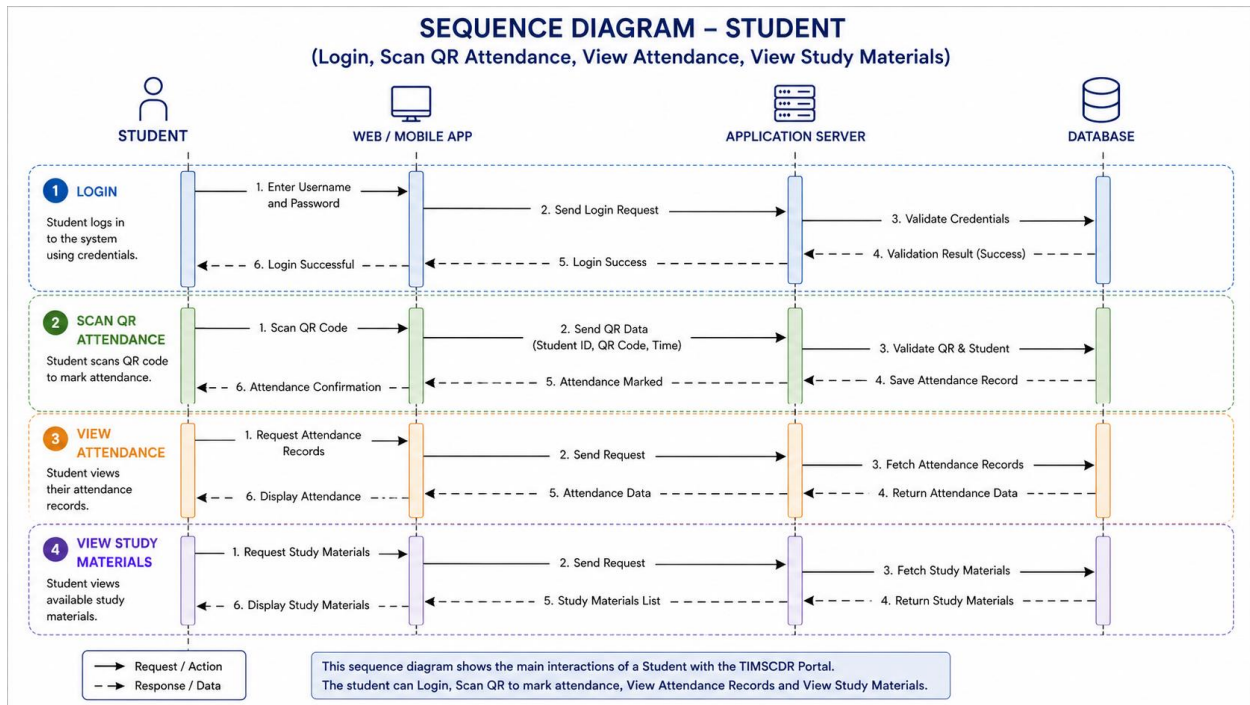
The Data Flow Diagram (DFD) represents how data flows between external entities (Student, Teacher, Admin) and the TIMSCDR Portal system. It shows how user inputs are processed by the system, stored in the database, and returned as meaningful outputs such as attendance records, reports, and study material.



3.7 Sequence Diagram

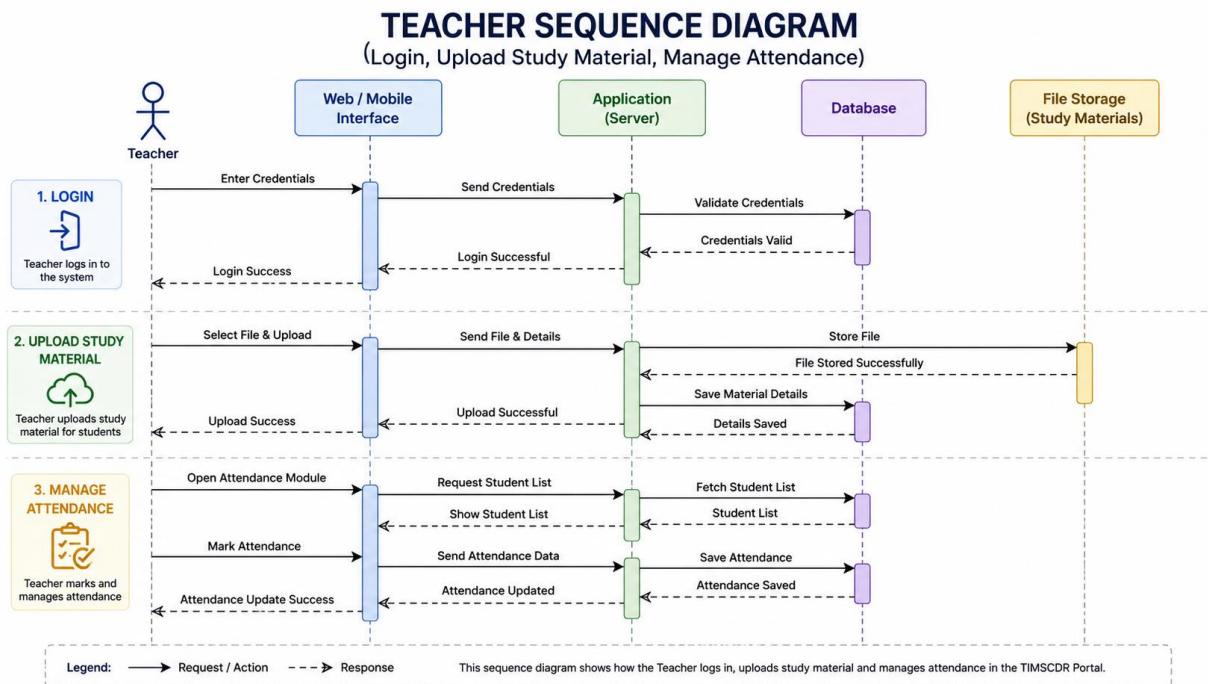
Sequence diagram for student

The sequence diagram illustrates the step-by-step interaction between the student and the TIMSCDR Portal system. It shows how the student logs in, marks attendance using QR code, views attendance records, and accesses study materials through system processes.



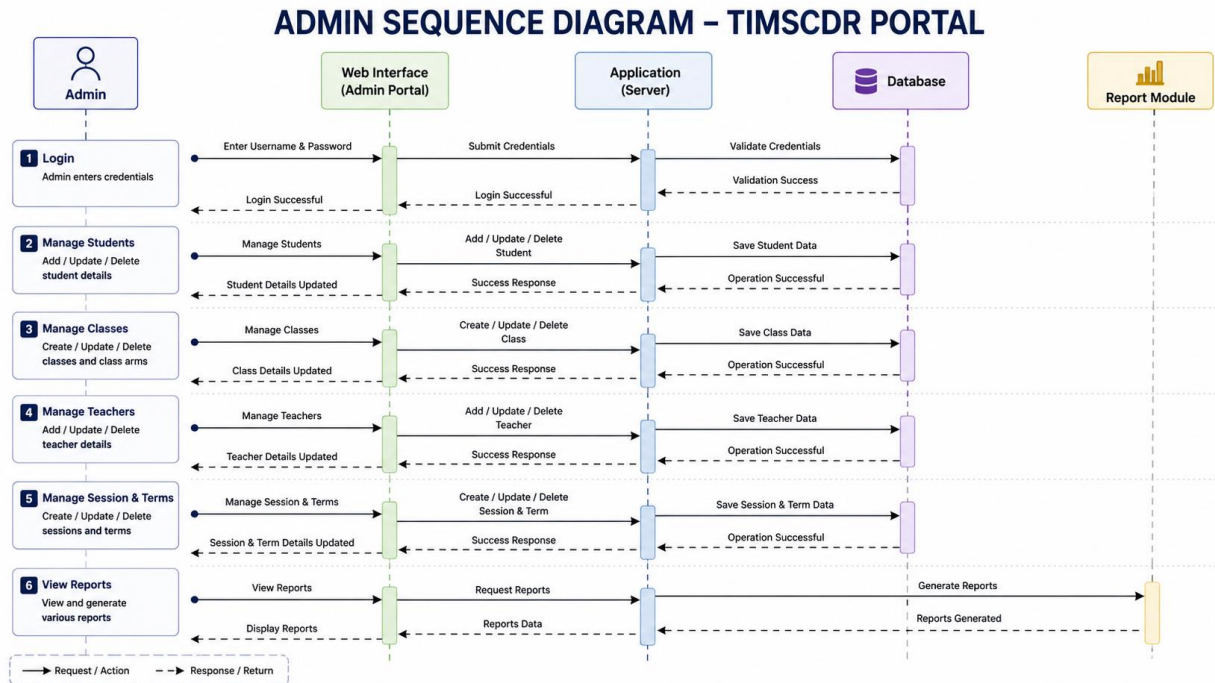
Sequence diagram for Teacher

The sequence diagram illustrates the interaction between the teacher, system, server, and database while performing operations such as login, uploading study materials, and managing attendance. It shows the step-by-step flow of data and responses within the TIMSCDR Portal.



Sequence diagram for Admin

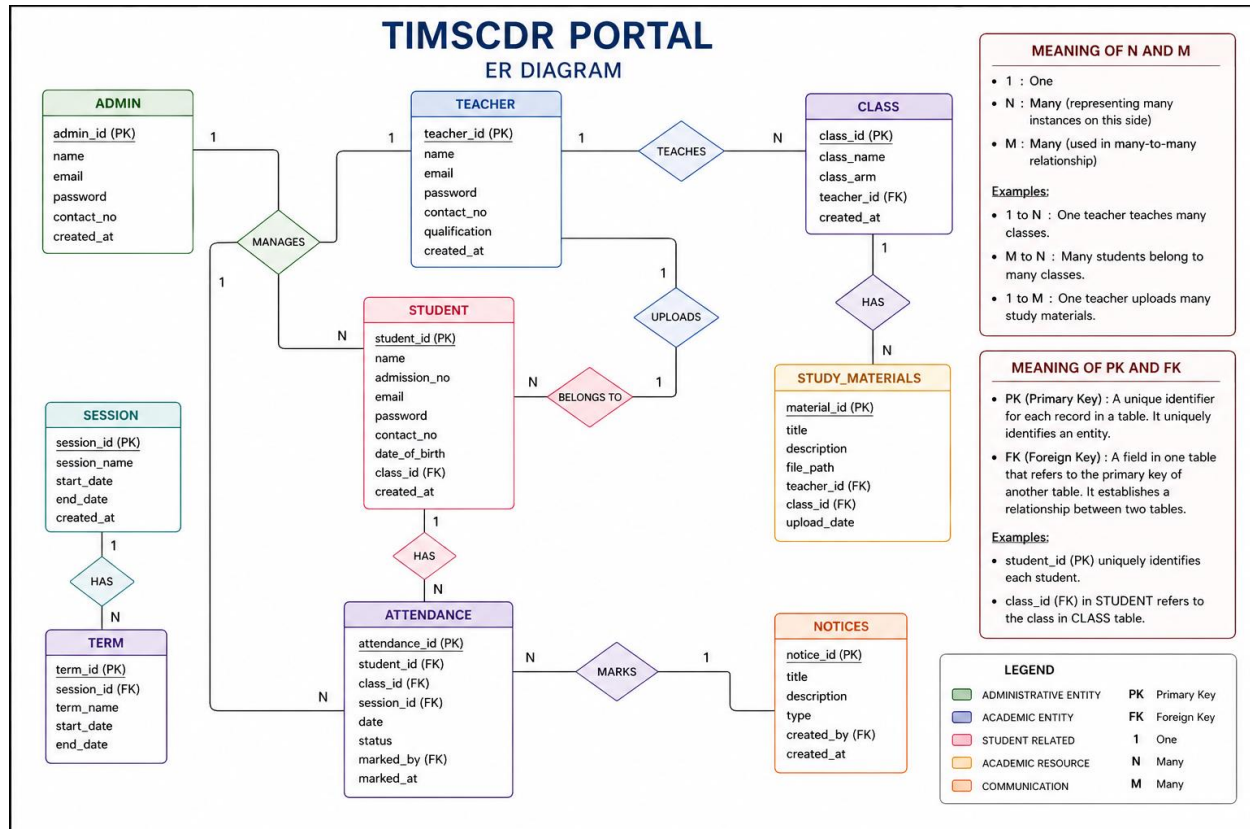
The admin sequence diagram represents the interaction between the admin, system, server, and database while performing administrative tasks such as managing students, classes, teachers, sessions, and reports. It shows the step-by-step flow of requests and responses required to control and manage the overall system.



ER diagram

The TIMSCDR Portal is a web-based academic management system that enables students, teachers, and administrators to manage attendance, study materials, and academic records efficiently.

It uses QR-based attendance and a centralized database to ensure accurate data management, secure access, and smooth communication between all users.



Description of Entities – TIMSCDR Portal

Entities represent the main users or components that interact with the system and exchange data. In the TIMSCDR Portal, the primary entities are Student, Teacher, Admin, and Database.

1. Student

The student is the primary user of the system who accesses academic services through the portal. The student logs in using credentials, marks attendance using QR code, views attendance records, and accesses study materials.

The student can also view profile details and track academic progress.

2. Teacher

The teacher manages classroom-related activities in the system.

The teacher logs in to upload study materials, manage attendance records, and monitor student performance.

The teacher also interacts with the system to update and maintain academic data.

3. Admin

The admin is responsible for controlling and managing the entire system.

The admin manages students, teachers, classes, sessions, and study materials.

The admin also generates reports and monitors overall system performance.

4. Database

The database stores all system data such as student records, teacher details, attendance, classes, study materials, and reports.

It ensures data consistency, security, and quick retrieval for system operations.

Chapter 4

Testing and Validation

Testing and validation are important phases of software development that ensure the system works correctly and meets user requirements. In the **TIMSCDR Portal**, testing was performed to verify that each module such as student login, QR-based attendance, study materials, and admin management functions properly and produces accurate results.

4.1 Code Efficiency

Code efficiency refers to how well the system performs using minimum resources such as time and memory. In the TIMSCDR Portal, efficient coding practices were followed to ensure smooth system performance.

Optimized database queries were used to fetch only required data, reducing load on the system. Input validation was implemented to prevent invalid login attempts and incorrect data entry. The system follows modular programming, where different modules like attendance, user management, and study materials are handled separately, improving maintainability and reducing code duplication.

Additionally, QR-based attendance processing is designed to be fast and time-bound (valid for limited seconds), ensuring quick response and real-time updates in attendance percentage.

4.2 Testing Approach

A systematic testing approach was adopted to identify and fix errors during different stages of development. Testing was performed both during and after the implementation of modules such as student, teacher, and admin functionalities.

Manual testing and functional testing techniques were used to ensure system reliability. Test cases were created based on real user scenarios such as login authentication, QR attendance scanning, attendance validation, study material upload, and report generation.

The system was tested under different conditions such as valid and invalid login credentials, expired QR codes, and incorrect data inputs to ensure proper error handling and system stability.

Unit Testing

Unit testing focuses on testing individual modules or components of the system independently. In the **TIMSCDR Portal**, each module such as login, QR attendance, study materials, and admin management was tested separately to verify its functionality.

Modules like **Student Login, QR Attendance, Teacher Upload, Admin Management, and Report Generation** were tested using both valid and invalid inputs. This helped in identifying errors early and ensured that each module works correctly and efficiently.

Test Case ID	Module Name	Test Description	Input Data	Expected Result	Actual Result	Status
UT-01	Student Login	Verify login with valid credentials	Valid ID & Password	Login successful	As expected	Pass
UT-02	Student Login	Verify login with invalid credentials	Wrong Password	Error message displayed	As expected	Pass
UT-03	QR Attendance	Verify QR code generation	Click attendance	QR generated (valid 30 sec)	As expected	Pass
UT-04	QR Attendance	Verify QR validation	Scan valid QR	Attendance marked	As expected	Pass

UT-05	QR Attendance	Verify expired QR	Scan after expiry	Error message shown	As expected	Pass
UT-06	Attendance Record	Verify attendance storage	Valid scan	Data stored in database	As expected	Pass
UT-07	Attendance View	Verify attendance percentage	Request attendance	% displayed correctly	As expected	Pass
UT-08	Teacher Module	Verify study material upload	Valid file upload	File uploaded successfully	As expected	Pass
UT-09	Admin Module	Verify student management	Add/Edit student	Data updated successfully	As expected	Pass
UT-10	Admin Module	Verify class management	Create class	Class created	As expected	Pass
UT-11	Admin Module	Verify session & term	Add session	Session saved	As expected	Pass
UT-12	Reports	Verify report generation	Request report	Report generated	As expected	Pass

Integration Testing

Integration testing focuses on testing how different modules of the system work together as a complete system. In the **TIMSCDR Portal**, integration testing was performed to ensure smooth interaction between modules such as Student, Teacher, Admin, Attendance, and Database.

The testing verified that data flows correctly between components. For example, when a student marks attendance using a QR code, the data is validated by the system, stored in the database, and reflected in attendance reports. Similarly, study materials uploaded by teachers are correctly accessible to students, and admin updates are reflected across the system.

This testing ensures that all modules are properly connected and function together without errors.

Test Case ID	Modules Involved	Test Description	Input Data	Expected Result	Actual Result	Status
IT-01	Student + Login System	Verify login integration	Valid credentials	Dashboard displayed	As expected	Pass
IT-02	Student + QR + Database	Verify QR attendance flow	Scan QR	Attendance stored in DB	As expected	Pass
IT-03	Student + Attendance Module	Verify attendance update	Mark attendance	% updated correctly	As expected	Pass
IT-04	Teacher + Study Material	Verify upload & access	Upload file	Students can view file	As expected	Pass

IT-05	Teacher + Attendance	Verify attendance management	Update attendance	Data updated in system	As expected	Pass
IT-06	Admin + Student Module	Verify student management	Add/Edit student	Changes reflected system-wide	As expected	Pass
IT-07	Admin + Class Module	Verify class creation	Create class	Class available to users	As expected	Pass
IT-08	Admin + Session Module	Verify session & term	Add session	Session applied to system	As expected	Pass
IT-09	Admin + Reports + Database	Verify report generation	Generate report	Accurate report displayed	As expected	Pass
IT-10	Full System Integration	Verify end-to-end flow	Login → Attendance → Report	Complete flow works correctly	As expected	Pass

Chapter 5

User Manual

1. User Documentation

The TIMSCDR Portal provides a simple and user-friendly interface for students, teachers, and administrators. The system is designed in such a way that users can easily access different functionalities based on their roles.

Student Module

Students can log into the system using their registered email and password. After successful login, students are redirected to the student dashboard where they can:

- Scan QR codes to mark attendance
- View attendance records and percentage
- Access study materials uploaded by teachers
- Update profile information
- View notifications and academic updates

The QR attendance feature allows students to mark attendance in real time using their mobile device camera through the web application.

Teacher Module

Teachers can securely log into the portal and perform academic-related tasks. The teacher dashboard provides functionalities such as:

- Uploading study materials and assignments
- Managing student attendance
- Monitoring attendance reports
- Viewing student information

Teachers can generate QR-based attendance sessions for each lecture, ensuring quick and accurate attendance management.

Admin Module

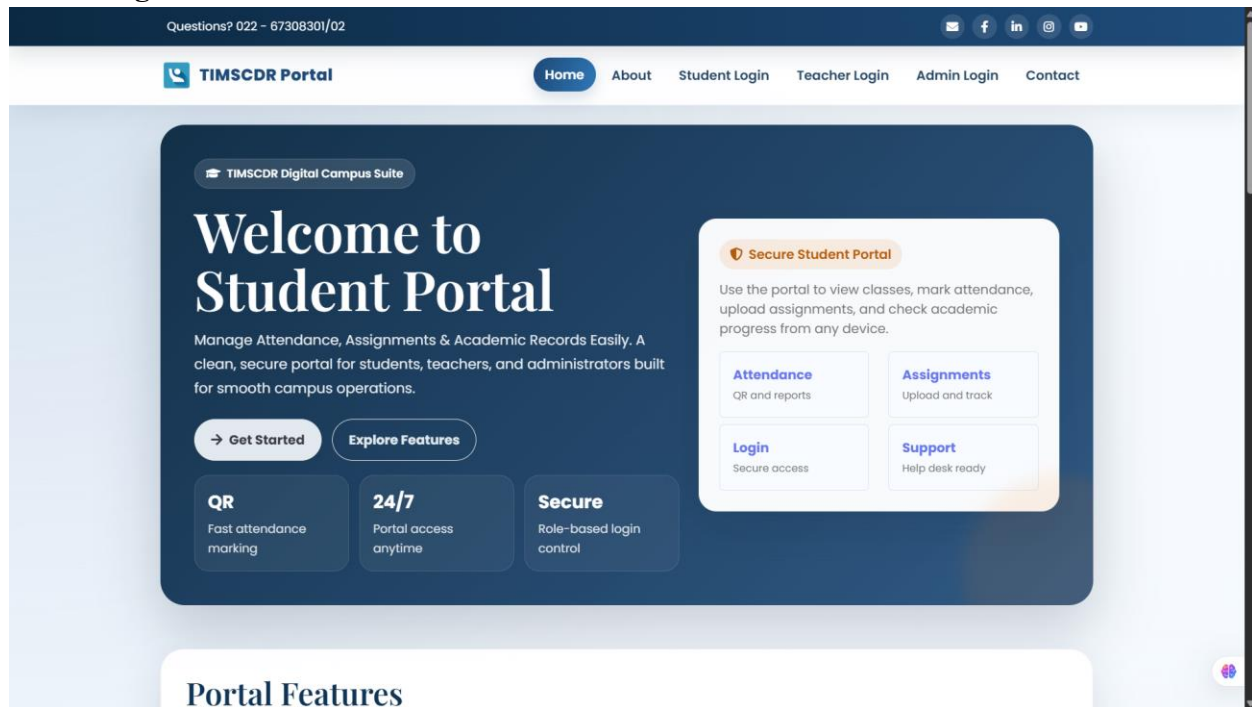
The administrator has complete control over the system. The admin dashboard allows:

- Managing students and teachers
- Creating and managing classes
- Managing academic sessions and terms
- Viewing attendance reports and system statistics
- Monitoring system activities

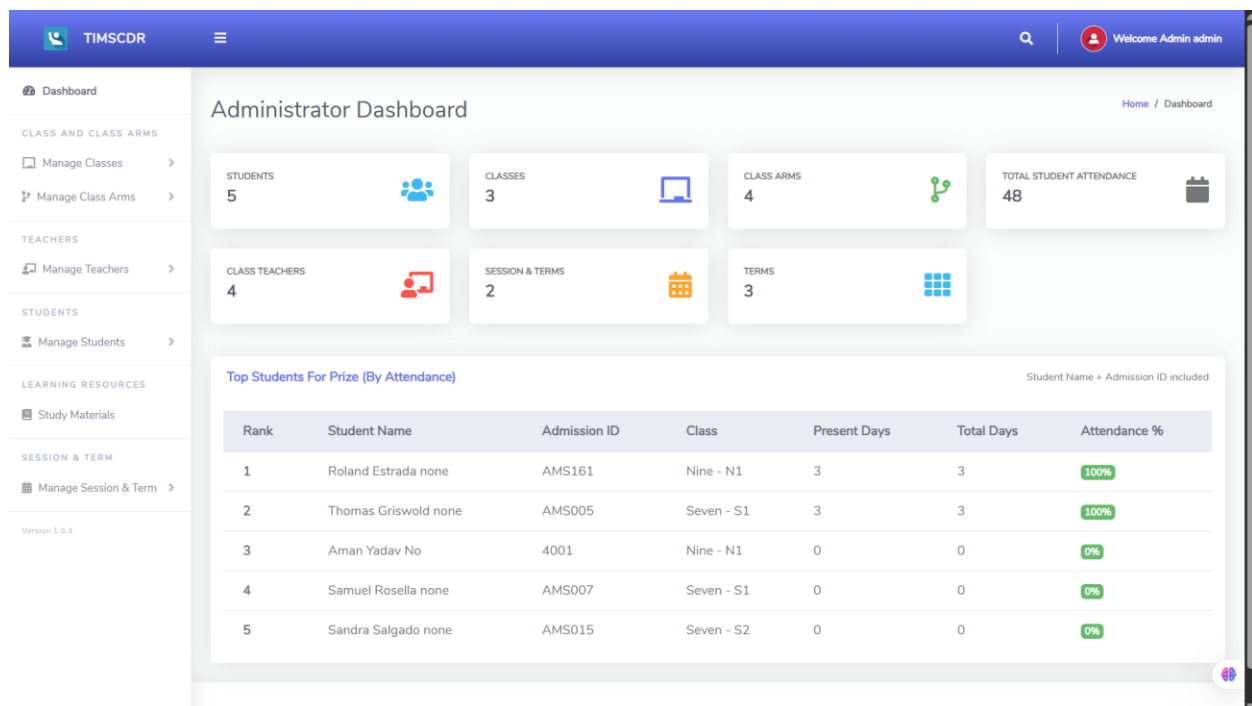
The admin can also manage system configurations and ensure smooth operation of all modules.

2. Output

Home Page



Admin Dashboard



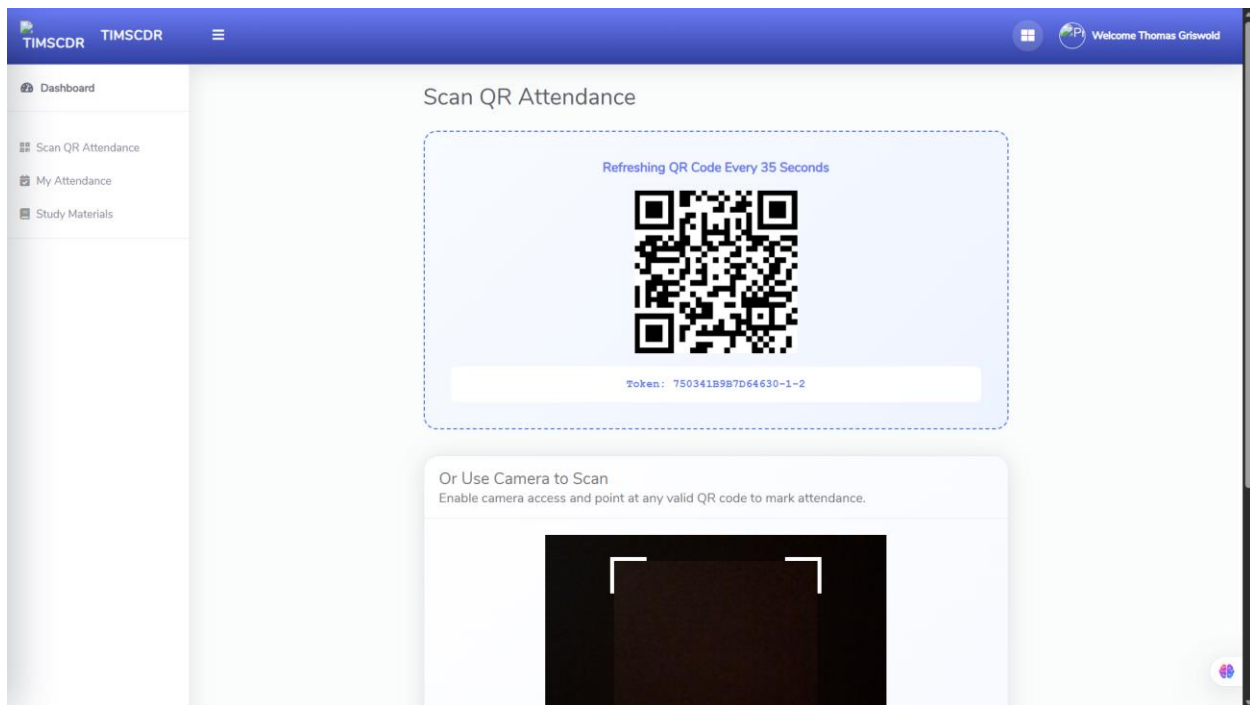
Student Dashboards

The screenshot shows the TIMSCDR Student Dashboard for Thomas Griswold. The top navigation bar includes the TIMSCDR logo and a welcome message. The left sidebar lists navigation options: Dashboard, Scan QR Attendance, My Attendance, and Study Materials. The main content area features a welcome message, a status bar indicating no subjects are running, and a grid of interactive cards for Scan QR, My Attendance, Study Materials, and My Profile. Below these are four summary cards: CLASS INFORMATION (Seven), DAYS PRESENT (3), ATTENDANCE RATE (100%), and STATUS (Active).

localhost/attendance-php/Student/studyMaterials.php

This screenshot shows the TIMSCDR Student Portal with the sidebar menu open. The user is Thomas Griswold (Admission No: AMS005). The sidebar menu is divided into two sections: CORE and STUDENT SERVICES. The CORE section includes Home, Leaderboard, Online Courses, Value Added Courses, Modules / Online Exam, and Academic Planning. The STUDENT SERVICES section includes Profile and Inventory. The background shows the same dashboard content as the previous screenshot, but it is dimmed.

Attendance Scanner



3. Code Snippets

Database Connection Code

```
<?php
$conn = new mysqli("localhost", "root", "", "timscdr_portal");

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
?>
```

Student Login Verification

```
<?php
session_start();

$email = $_POST['email'];
$password = $_POST['password'];
```

```

$query = "SELECT * FROM users WHERE email='$email'";
$result = mysqli_query($conn, $query);

$user = mysqli_fetch_assoc($result);

if(password_verify($password, $user['password'])) {
    $_SESSION['user_id'] = $user['id'];
    header("Location: dashboard.php");
}
else {
    echo "Invalid Credentials";
}
?>

```

QR Attendance Scan Logic

```

<?php
$student_id = $_SESSION['user_id'];
$subject = $_POST['subject'];

$date = date("Y-m-d");

$query = "INSERT INTO attendance(student_id, subject, date, status)
VALUES('$student_id', '$subject', '$date', 'Present')";

mysqli_query($conn, $query);
?>

```

QR Scanner JavaScript

```

<script src="https://unpkg.com/html5-qrcode"></script>

<script>
function onScanSuccess(decodedText) {
    fetch("mark_attendance.php", {
        method: "POST",
        body: new URLSearchParams({
            qr: decodedText
        })
    });
}

```

```
}  
  
new Html5QrcodeScanner("reader", {  
  fps: 10,  
  qrbox: 250  
}).render(onScanSuccess);  
</script>
```

Chapter 6

6.1 Conclusion

The **TIMSCDR Portal** is a web-based student management system developed to simplify academic processes such as attendance management, study material access, and user administration. The system successfully integrates multiple modules including student, teacher, and admin functionalities into a single platform.

One of the key features of the system is QR-based attendance, which ensures accuracy and reduces manual effort. The system provides a user-friendly interface, secure login, and real-time data updates, making it efficient and reliable for academic use.

Overall, the TIMSCDR Portal achieves its objective of digitizing academic activities, improving data management, and enhancing communication between users. It serves as a scalable and practical solution for modern educational institutions.

6.2 Limitations

Although the system performs efficiently, there are certain limitations:

- The system requires an active internet connection for proper functioning
- QR-based attendance depends on device compatibility and camera access
- Limited security features compared to enterprise-level systems
- No mobile application support currently
- Performance may be affected with a large number of users

6.3 Future Scope

The TIMSCDR Portal can be enhanced further with additional features and improvements:

- Development of a mobile application for better accessibility
- Integration with cloud technology for scalability and data backup
- Implementation of advanced security features such as encryption and OTP login
- AI-based analytics for student performance prediction
- Online examination and assignment submission system
- Integration with biometric attendance for higher accuracy.